Sajan Patel

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Experience

Software Developer I
May 2026 – Present

Iowa State University – Information Technology Services (ITS) *Ames, IA*

- Build and customize AI agents and chatbots using Microsoft Copilot Studio, integrating AI solutions into enterprise platforms including Microsoft Teams, SharePoint, and ServiceNow to enhance institutional services for faculty, staff, and students.
- Engineered a Power BI analytics dashboard secured with Row-Level Security (RLS) and embedded in SharePoint, giving employees self-service access to their individual AI usage metrics (credits, messages, tool calls) and reducing manual reporting requests.
- Automated deployment and low-code workflows using Power Automate and PowerShell, and implemented REST API connections to external endpoints to support new AI use cases.
- Translated stakeholder requirements into technical solutions, cleaned and preprocessed data with Python and SQL, and tracked development in JIRA within an Agile/Scrum delivery model.

Full Stack Software Engineer Intern
Dec. 2024 – Jul. 2025

Frigate Global *London, UK*

- Architected and deployed RESTful APIs using Node.js and Spring Boot, serving data to 3+ web and mobile clients with PostgreSQL.
- Built responsive front-end components with JavaScript and React, reducing mobile bounce rate by 20% through improved UI/UX.
- Collaborated with 8+ engineers and UI/UX designers in an Agile/Scrum environment, shipping features on two-week sprints.
- Participated in code reviews and pull request workflows to maintain code quality and team standards.

Software Engineering Intern
May 2022/23 – Aug. 2022/23

Marmon Foodservice Technologies *St. Charles, IL*

- Designed and implemented an automated CI/CD workflow, streamlining the team's build and deployment pipeline.
- Integrated third-party plugins and UI component libraries, enhancing the design pipeline and increasing team efficiency by 70%.
- Built comprehensive test suites using embedded, integration, and system testing, reducing QA execution time by 50%.

Projects

Chat-with-Documents AI App | *FastAPI, MongoDB, AWS S3, LangChain, OpenAI API, FAISS, Python*

- Built a full-stack AI application using FastAPI enabling users to upload documents and chat with content via GPT-powered conversational AI.
- Implemented vector-based semantic search with FAISS and OpenAI embeddings, achieving sub-second retrieval for context-aware Q&A.
- Integrated MongoDB for session-based conversation history and AWS S3 for scalable document storage.
- Designed a clean, user-friendly front-end interface for document upload and real-time chat interactions.

Education

Iowa State University
Ames, IA

B.S. in Software Engineering | GPA: 3.43 *Graduated May 2025*

Senior Project | CyVital Project

- Engineered a real-time Python application to interface with EMG, ECG, and pulse oximeter sensors via microcontroller, applying FFT and bandpass filtering for bio-signal acquisition and feature extraction.
- Developed an interactive GUI for visualizing physiological data, enhancing lab workflow efficiency by 40%.

Technical Skills

Languages: Python, Java, JavaScript, SQL, HTML/CSS, PowerShell, VBA
AI & Machine Learning: Machine Learning (supervised, unsupervised, reinforcement learning), Microsoft Copilot Studio, LangChain, OpenAI API, FAISS, prompt engineering
Microsoft & Automation: Power BI, Power Automate, SharePoint, Microsoft Teams, ServiceNow, Microsoft 365
Frameworks & Cloud: Node.js, Spring Boot, React, FastAPI, REST APIs, AWS (S3, EC2), Docker
Tools & Methods: PostgreSQL, MongoDB, Git/GitHub, CI/CD, JIRA, Agile/Scrum, VS Code